23/09/2026

Code-C_Phase-3

Aakash

Medical | IIT-JEE | Foundations

Corporate Office : AESL, 3rd Floor, Incuspaze Campus-2, Plot No. 13, Sector-18,
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MM : 720

Fortnightly Test for NEET-2027 (XII-Passed)_FT03C

Time : 180 Min.

Topics Covered:
Physics: Work, Energy & Power, System of Particles & Rotational Motion
Chemistry: Chemical Bonding and Molecular Structure, Thermodynamics
Botany: Biological Classification-II: (from Cyanobacteria to Lichens), Morphology of Flowering Plants
Zoology: Breathing & Exchange of Gases-II: (Upto disorders), Body Fluids & Circulation

General Instructions :
Duration of Test is 3 hrs.
The Test consists of 180 questions. The maximum marks are 720.
There are four parts in the question paper consisting of Physics, Chemistry, Botany and Zoology having 45 questions in each part of equal weightage.
Each question carries +4 marks. For every wrong response, –1 mark shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
Use blue/black ballpoint pen only to darken the appropriate circle.
Mark should be dark and completely fill the circle.
Dark only one circle for each entry.
Dark the circle in the space provided only.
Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

CC-003
@examlix

PHYSICS

1. Consider the following four statements.
   (a) Work done by friction is always negative
   (b) Work done by spring force is always negative
   (c) Gravitational force is a conservative force
   (d) Centre of mass of a body may lie outside the material of the body
   The incorrect statement(s) from the above are
   [✓ (1) (a), (b) and (d) only]
   (2) (a) and (b) only
   (3) (b) and (c) only
   (4) (b) and (d) only

2. A particle of mass *m* is attached with a vertical string of length *l* and is given a velocity at its lower most position such that it just complete the circle. On basis of given information match the column I with column II and choose the correct option

	Column I		Column II
a	Radial acceleration at lower most position (in m/s^2)	(i)	*g*
b.	Tension in string at horizontal position	(ii)	Zero
c.	Radial acceleration at upper most position (in m/s^2)	(iii)	3 *mg*
d.	Tension in string at uppermost position (in N)	(iv)	5 *g*

   (1) a(ii), b(i), c(iv), d(iii)
   (2) a(iv), b(iii), c(i), d(ii)
   (3) a(i), b(iv), c(iii), d(ii)
   [✓ (4) a(i), b(iii), c(ii), d(iv)]

3. Consider the following statement out of which one is labelled as assertion and other as reason.
   Assertion (A): For a particle moving in *xy* plane with uniform velocity (*v*), its angular momentum (*L*) about origin remains constant.
   Reason (R): In the above situation, the perpendicular distance from the origin to line of motion of particle is constant.
   In the light of above statements, choose the **correct** option.
   (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
   [✓ (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)]
   (3) (A) is true, (R) is false
   (4) Both (A) and (R) are false

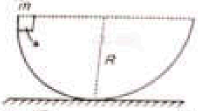
4. Given below are two statements:
   Statement I: Torque τ in rotational motion is analogous to force in translatory motion.
   Statement II: For a rotating body rate of change of angular momentum *L*, $\frac{dL}{dt} = I\omega$ (where *I* → moment of inertia, ω- angular speed)
   In the light of the given statements, choose the most appropriate answer from the options given below.
   (1) Both statement I and statement II are correct
   (2) Both statement I and statement II are incorrect
   (3) Statement I is correct but statement II is incorrect
   (4) Statement I is incorrect but statement II is correct

5. A particle is acted upon by a constant power. Then, which of the following physical quantity remains constant?
   (1) Speed
   (2) Rate of change of acceleration
   (3) Rate of change of kinetic energy
   (4) Kinetic energy

6. The collision in which the relative velocity of approach before collision is equal to relative velocity of separation after collision is
(1) Completely inelastic collision
(2) Elastic collision
(3) Inelastic collision
(4) Both elastic and inelastic collision

7. Consider a uniform rod of mass m and length l placed along y-axis. The rod has translational velocity $v\hat{i}$ and an angular velocity $\omega\hat{k}$ about its centre of mass (c.o.m). The angular momentum ($\vec{L}$) of the rod about it's c.o.m. will be (where $\hat{i}, \hat{j}, \hat{k}$ are unit vectors along x-axis, y-axis and z-axis respectively)
(1) $\left(\frac{mvl}{2} - \frac{m\omega l^2}{12}\right)\hat{k}$
(2) $\frac{m\omega l^2}{12}\hat{k}$
(3) $\left(\frac{mvl}{2} + \frac{m\omega l^2}{12}\right)(-\hat{k})$
(4) $\frac{m\omega l^2}{12}(-\hat{k})$

8. A mass m slips along the smooth wall of a hemispherical surface of radius R. The velocity at the bottom of the surface is



(1) $\sqrt{Rg}$
(2) $\sqrt{2Rg}$
(3) $2\sqrt{\pi Rg}$
(4) $\sqrt{\pi Rg}$

A force $\left(4\hat{i} + \hat{j} - 2\hat{k}\right)$ N is acting on a body moving with
instantaneous velocity $\left(2\hat{i} - 2\hat{j} + 3\hat{k}\right)$ m/s. The power exerted
by the force is
(1) 4 W
(2) 5 W
(3) 2 W
(4) 8 W

When a body moves with a constant speed along a circle
(1) No work is done on it
(2) No acceleration is produced in the body
(3) No force acts on the body
(4) Its velocity remains constant

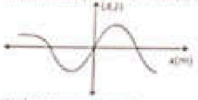
Non conservative force among the following is
(1) Gravitational force
(2) Friction
(3) Spring force
(4) Electrostatic force

Two bodies of masses 3m and 4m have their kinetic energy
in the ratio 3 : 1. The ratio of their momenta is
(1) 2 : 1
(2) 3 : 2
(3) 3 : 1
(4) 4 : 3

A particle moves on x-axis such that its energy varies as a
function, E (in J) $= 4t^3$, where t is in second. The average
rate of energy of the particle in $t = 0$ to $t = 2$ s is
(1) J
(2) 8 J
(3) J
(4) 32 J

14. **Assertion (A):** Work done on a body A by another body B is not necessarily equal and opposite to the work done on body B by A.
Reason (R): Force on a body A exerted by the body B is equal and opposite to that on body B by body A.
In the light of above statements, choose the correct option.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) Both (A) and (R) are false

15. Variation of potential energy of a particle in space is represented in the following graph. If the particle is released from rest at origin, then it will start to move



(1) Along negative x-axis
(2) Along positive x-axis
(3) The particle will stay at origin
(4) Either (1) or (2)

16. A cord is used to lower vertically a block of mass m through a distance d at a constant downward acceleration of $\frac{g}{4}$. Work done by the cord on the block is
(1) $\frac{3mgd}{4}$
(2) $\frac{mgd}{4}$
(3) mgd
(4) $-\frac{3mgd}{4}$

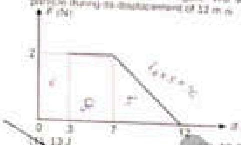
17. Two springs A and B having force constant k_A and k_B ($k_A = 3k_B$) are stretched by applying equal forces. If energy stored in A is E, then that in B is
(1) $\frac{E}{3}$
(2) E
(3) $3E$
(4) $9E$

18. A ball is dropped from height h on a horizontal floor goes up to height $\frac{h}{2}$ after hitting the floor. The percentage loss of energy in the impact is
(1) 50%
(2) 75%
(3) 25%
(4) 0%

19. In all type of collisions, which of the following quantity is conserved?
(1) Kinetic energy
(2) Mechanical energy
(3) Linear momentum
(4) All of these

20. A box is dragged across a floor by a rope which makes an angle 60° with horizontal. The tension in the rope is 50 N while the box is dragged for 10 m. The work done by tension is
(1) 500 J
(2) 750 J
(3) 250 J
(4) $250\sqrt{3}$ J

21. Force F on a particle moving in a straight line varies with distance d as shown in the figure. The work done on the particle during its displacement of 12 m is

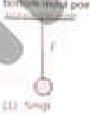


(1) 13 J
(2) 18 J
(3) 21 J
(4) 26 J

22. The potential energy of a particle in a force field is $U = \frac{A}{r^2} - \frac{B}{r}$, where A and B are positive constants and r is the distance of particle from the centre of the field. For stable equilibrium, the distance of the particle is
(1) $\frac{A}{B}$
(2) $\frac{B}{A}$
(3) $\frac{B}{2A}$
(4) $\frac{2A}{B}$

23. During head on elastic collision two bodies of mass m_1 and m_2 exchange their velocities when
(1) $m_1 = 2m_2$
(2) $m_1 = m_2$
(3) $m_1 = 4m_2$
(4) $m_1 = 3m_2$

24. A ball of mass m is connected to a fixed point with light string of length l. The minimum kinetic energy required at the bottom most point to complete a circle is



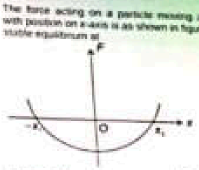
(1) 5mgl
(2) $\frac{5mgl}{4}$
(3) $\frac{1}{2}mgl$
(4) $\frac{5mgl}{2}$

25. The quantity whose unit is watt-second, is
(1) Power
(2) Energy
(3) Force
(4) Velocity

26. A body of mass 2 kg starts from rest and reaches a speed of 3 ms^{-1}. The work done by the forces in this act is
(1) 4 J
(2) 6 J
(3) 9 J
(4) 10 J

27. A body with mass 1 kg moves with velocity $\vec{v} = \left(2\hat{i} + 4\hat{j}\right)$ ms^{-1}. Then its kinetic energy is
(1) 8 J
(2) 12 J
(3) 10 J
(4) 6 J

28. The force acting on a particle moving along x-axis varies with position on x-axis is as shown in figure. The particle is at stable equilibrium at



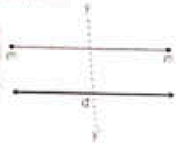
(1) $x = -x_1$
(2) $x = x_1$
(3) $x = 0$
(4) Both (1) and (2)

$4t - 3t^2 + t^3$
$\frac{d\theta}{dt} = 4 - 6t + 3t^2$
[illegible]

29. The angular position of a point on the rim of rotating wheel is given by $\theta = (4t - 3t^2 + t^3)$, where θ is in radians and t is in seconds. What is the average angular acceleration for the time interval from $t = 2$ s to $t = 4$ s?
(1) 12 rad/s^2
(2) 10 rad/s^2
(3) 2 rad/s^2
(4) 1 rad/s^2

30. A disc is rotating with angular speed ω. If a child sits gently on it, then angular momentum of system
(1) Decreases
(2) Increases
(3) Remains constant
(4) Is zero

31. Find moment of inertia about an axis yy' which passes through the centre of mass of the two particle system as shown



(1) $\frac{md^2}{4}$
(2) md^2
(3) $\frac{md^2}{2}$
(4) $\frac{md^2}{8}$

32. A particle of mass 5 g is moving with uniform speed of $3\sqrt{2}$ cm/s in the $x - y$ plane along the line $y = x + 4$. The magnitude of its angular momentum about the origin in g cm^2/s is (x and y are in cm)
(1) Zero
(2) 30
(3) 60
(4) $3\sqrt{2}$

33. A torque of 20 N m is applied on a wheel for 2 second. The change in angular momentum of wheel is
(1) 20 kg $\frac{m^2}{s}$
(2) 40 kg $\frac{m^2}{s}$
(3) 5 kg $\frac{m^2}{s}$
(4) 10 kg $\frac{m^2}{s}$

34. The angular velocity of flywheel changes from 60 rpm to 120 rpm. Moment of inertia of flywheel is $\frac{1}{\pi}$ kg m^2. Change in angular momentum is (in SI unit)
(1) $\frac{3\pi}{2}$
(2) $\frac{3}{\pi}$
(3) $\frac{2\pi}{3}$
(4) $\frac{\pi}{2}$

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6. The collision in which the relative velocity of approach before collision is equal to relative velocity of separation after collision is
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7. Consider a uniform rod of mass m and length l placed along x-axis. The rod has translational velocity $v\hat{i}$ and an angular velocity $\omega\hat{k}$ about its centre of mass (c.o.m). The angular momentum $(\vec{L})$ of the rod about it's c.o.m. will be (where $\hat{i}$, $\hat{j}$, $\hat{k}$ are unit vectors along x-axis, y-axis and z-axis respectively)
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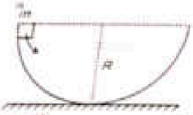

(1) $\sqrt{Rg}$
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9. A force $(4\hat{i}+\hat{j}-2\hat{k})$ N is acting on a body moving with instantaneous velocity $(2\hat{i}+2\hat{j}+3\hat{k})$ m/s. The power exerted by the force is
(1) 4 W
(2) 5 W
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10. When a body moves with a constant speed along a circle
(1) No work is done on it
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11. Non conservative force among the following is
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12. Two bodies of masses 3m and 4m have their kinetic energy in the ratio 3 : 1. The ratio of their momenta is
(1) 2 : 1
(2) 3 : 2
(3) 3 : 1
(4) 4 : 3

13. A particle moves on x-axis such that its energy varies as a relation, E (in J) $= 4t^3$ where t is in second. The average value of energy of the particle in $t = 0$ to $t = 2$ s is
(1) 4 J
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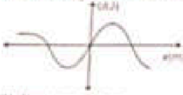

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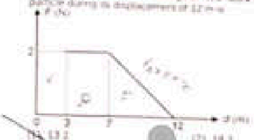

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(1) $\frac{B}{A}$
(2) $\frac{A}{B}$
(3) $\frac{B}{2A}$
(4) $\frac{2A}{B}$

23. During head on elastic collision between two bodies of mass m_1 and m_2 they exchange their velocities when
(1) $m_1 > m_2$
(2) $m_1 = m_2$
(3) $m_1 < m_2$
(4) $m_1 = 3m_2$

24. A ball of mass m connected to a fixed point with light string of length l. The minimum kinetic energy required at the bottom most point to complete a circle is

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25. The quantity whose unit is watt second, is
(1) Power
(2) Energy
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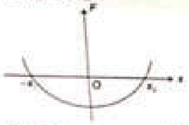

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(3) $x = 0$
(4) Both (1) and (2)

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(1) Decreases
(2) Increases
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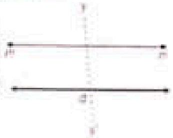
(4) Is zero

31. Find moment of inertia about an axis yy' which passes through the centre of mass of the two particle system as shown



(1) $\frac{md^2}{4}$
(2) md^2
(3) $\frac{md^2}{2}$
(4) $\frac{3md^2}{4}$

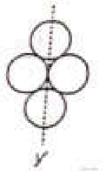
32. A particle of mass 5 g is moving with uniform speed of $3\sqrt{2}$ cm/s in the $x-y$ plane along the line $y = x + 4$. The magnitude of its angular momentum about the origin in g cm^2/s is (x and y are in cm)
(1) Zero
(2) 30
(3) 60
(4) $3\sqrt{2}$

33. A torque of 20 N m is applied on a wheel for 2 second. The change in angular momentum of wheel is
(1) 20 kg$\frac{m^2}{s}$
(2) 40 kg$\frac{m^2}{s}$
(3) 4 kg$\frac{m^2}{s}$
(4) 10 kg$\frac{m^2}{s}$

34. The angular velocity of flywheel changes from 60 rpm to 120 rpm. Moment of inertia of flywheel is $\frac{1}{\pi}$ kg m^2. Change in angular momentum is (in SI unit)
(1) [illegible]
(2) [illegible]
(3) [illegible]
(4) [illegible]

35. A thin wire of mass [illegible] and length L is bent to form a circular ring. The moment of inertia of ring about an axis passing through the centre and perpendicular to the plane is
(1) [illegible]
(2) [illegible]
(3) [illegible]
(4) [illegible]

36. Four discs each of mass M and radius R are arranged as shown in the figure. The moment of inertia of the system about axis yy' will be



4x+8+18 = 0
−28 = 4x

(1) 3 MR^2
(2) [illegible] MR^2
(3) 5 MR^2
(4) [illegible] MR^2

37. A force $\vec{F} = \alpha\hat{i} - 2\hat{j} + 3\hat{k}$ is acting on a particle at the point $\vec{r} = 4\hat{i} - 4\hat{j} + 6\hat{k}$. The value of α for which angular momentum of particle about origin is conserved, is
(1) 1
(2) 2
(3) −2
(4) 4

38. A constant torque 10 N-m applied on a body at some instant of time. Body is rotating with angular velocity 10 rad/s. The instantaneous power of the body is
(1) 100 W
(2) 10 W
(3) 20 W
(4) Zero

39. Which of the following relation is correct?
(1) $\vec{F} = \vec{r} \times \vec{\tau}$
(2) $\vec{\tau} = \vec{r} \times \vec{F}$
(3) $\vec{F} = \vec{\tau} \times \vec{r}$
(4) $\vec{\tau} = \vec{F} \times \vec{r}$

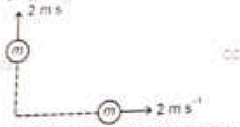
40. A disc starts rotating with constant angular acceleration of π rad/s² about a fixed axis perpendicular to its plane and passing through its centre. The angular velocity of the disc at t = 3 s
(1) 2π rad/s
(2) 3π rad/s
(3) 4π rad/s
(4) 5π rad/s

41. The value of the cross product of vectors $(2\hat{i} - 4\hat{j})$ and $(\hat{i} - 2\hat{j})$ is
(1) $-2\hat{i} + 4\hat{k}$
(2) $2\hat{i} - 4\hat{k}$
(3) $-4\hat{k}$
(4) Zero

42. A ring of mass 4 kg rotates about its diameter with angular speed 1000 rad/s. If ring has radius 0.25 m, then magnitude of angular momentum of ring about this axis is
(1) 62.5 J s
(2) 125 J s
(3) 250 J s
(4) 500 J s

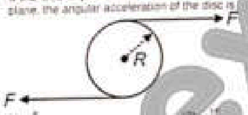
43. Two particles each of mass 1 kg are located at the point having co-ordinates (1, 0, 1) and (1, 1, 0). The co-ordinates of the centre of mass will be
(1) $(\frac{1}{2}, \frac{1}{2}, 1)$
(2) $(1, \frac{1}{2}, \frac{1}{2})$
(3) (1, 1, 1)
(4) $(\frac{1}{2}, 1, 0)$

44. Two balls each of mass m are moving in a horizontal plane (x − y) as shown.



The speed of centre of mass of the system is
(1) Zero
(2) $2\sqrt{2}$ m s⁻¹
(3) $\sqrt{2}$ m s⁻¹
(4) $\frac{1}{\sqrt{2}}$ m s⁻¹

45. Two equal and opposite forces are applied tangentially to a uniform disc of mass M and radius R as shown in the figure. If the disc is pivoted at its centre and free to rotate in its plane, the angular acceleration of the disc is



(1) $\frac{F}{MR}$
(2) [illegible]
(3) $\frac{2F}{MR}$
(4) [illegible]

CHEMISTRY

46. Consider the following processes
(a) Boiling of egg
(b) $H_2(g) \rightarrow 2H(g)$
(c) Stretching of rubber
The entropy increases in
(1) b only
(2) a and b only
(3) b and c only
(4) a and c only

47. If a system absorb 100 J of heat and does 60 J of work, then the internal energy change for the system is
(1) 160 J
(2) − 40 J
(3) − 160 J
(4) 40 J

48. If $\Delta_f H^\circ$ of $CaCO_3(s)$, CaO(s) and $CO_2(g)$ are −1200, −630 and −400 kJ mol⁻¹ respectively then $\Delta_r H^\circ$ of the reaction $CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$ is
(1) −170 kJ mol⁻¹
(2) 170 kJ mol⁻¹
(3) −2230 kJ mol⁻¹
(4) 2230 kJ mol⁻¹

49. Identify the correct order against the property mentioned
(a) C – Cl > N – O > C – H : Covalent bond length
(b) $N_2 > H_2 > HCl$: Bond enthalpy
(c) $O_2^- > N_2 > F_2$: Bond order
(d) $BCl_3 > NH_3 > H_2O$: Bond angle
choose the correct option
(1) (a), (b) and (d) only
(2) (a) and (d) only
(3) (c) and (d) only
(4) (a) and (c) only

50. Consider the following statements
a. o-nitrophenol forms intramolecular hydrogen bond
b. Boiling point of H_2S is greater than H_2O
c. Hydrogen bond is stronger than covalent bond
The incorrect statement(s) are
(1) a only
(2) a and c only
(3) a and b only
(4) b and c only

51. Entropy change for the vaporisation of 9 g water at 373 K and 1 atm is (Enthalpy of vaporisation of water is 40.66 kJ/mol)
(1) 109.2 JK⁻¹
(2) 54.5 JK⁻¹
(3) 27.25 JK⁻¹
(4) 38.5 JK⁻¹

52. The correct relation between molar heat capacities at constant pressure (C_P) and at constant volume (C_V) is
(1) $C_P + C_V = R$
(2) $C_P - C_V = 2R$
(3) $C_P - C_V = R$
(4) $C_P \cdot C_V = \frac{R}{2}$

53. If the observed dipole moment of AB molecule is 2.5 D and A–B bond length is 125 pm then percentage ionic character of the molecule AB will be
(1) 35.2%
(2) 41.6%
(3) 28.6%
(4) 22.6%

54. Match List I with List II

	List-I (Process)		List-II (Conditions)
a.	Isothermal process	(i)	Carried out at constant entropy
b.	Isochoric process	(ii)	Carried out at constant temperature
c.	Isobaric process	(iii)	Carried out at constant volume
d.	Isentropic process	(iv)	Carried out at constant pressure

Choose the correct answer from the options given below
(1) a(iv), b(iii), c(ii), d(i)
(2) a(iv), b(ii), c(iii), d(i)
(3) a(i), b(ii), c(iii), d(iv)
(4) a(ii), b(iii), c(iv), d(i)

55. Five litres of an ideal gas at a pressure of 10 atm expands isothermally at 25°C into vacuum until its total volume is 25 litres. Then work done in the expansion will be
(1) −10941 J
(2) −9970.5 J
(3) +10941 J
(4) Zero

56. Bomb-calorimeter is used to experimentally measure
(1) ΔH
(2) ΔU
(3) ΔS
(4) ΔG

57. The highest occupied molecular orbital for Ne_2 is
(1) σ^*_{2p}
(2) π_{2p}
(3) σ_{2p}
(4) π^*_{2p}

58. For a reaction
$X_2 + Y_2 \rightarrow 2XY$; ΔH = −400 kJ
If bond energy of X_2, Y_2 and XY are in ratio of 1 : 2 : 3 respectively then bond energy of Y_2 in kJ mol⁻¹ will be
(1) 400
(2) 200
(3) $\frac{400}{3}$
(4) $\frac{800}{3}$

59. Consider the following statements about PCl_5
(a) It has sp^3d hybridisation.
(b) Equatorial bonds are shorter than axial bonds.
(c) Maximum number of atoms lying in one plane is 5.
The correct statements are
(1) (a) and (b) only
(2) (b) and (c) only
(3) (a) and (c) only
(4) (a), (b) and (c)

60. Which among the following is not a state function?
(1) Pressure
(2) Volume
(3) Internal energy
(4) Heat

61. For the reaction, A(g) + B(s) → AC(g); ΔH° = −10 kJ and ΔS° = −40 J K⁻¹, then value of ΔG° at 298 K will be
(1) 2.98 kJ
(2) 1.92 kJ
(3) 3.94 kJ
(4) 0.72 kJ

62. Which among the following is an intensive property?
(1) Mass
(2) Volume
(3) Entropy
(4) Pressure

63. 10 moles of an ideal gas at 27°C expands isothermally and reversibly from a volume of 12 L to 240 L. The work done in the process is
(1) −74.73 kJ
(2) −89.28 kJ
(3) −37.14 kJ
(4) −57.2 kJ

64. Equilibrium constant for a reaction with standard gibbs free energy of −4.2 kcal at 127°C will be
(1) 10^4
(2) 10^2
(3) 50
(4) 500

65. Covalent character is maximum in the following compound
(1) CaF_2
(2) $CaCl_2$
(3) $CaBr_2$
(4) CaI_2

66. The paramagnetic species among the following is
(1) N_2
(2) O_2
(3) B_2
(4) H_2

67. If enthalpy of combustion of carbon (graphite) to CO_2 (g) is −393 kJ mol⁻¹ then the heat released upon formation of [illegible] g of CO_2 (g) from carbon (graphite) and oxygen gas is
(1) 283.52 kJ
(2) 245.63 kJ
(3) 323.74 kJ
(4) 196.63 kJ

68. Which of the following set of acid and base will give maximum heat of neutralisation?
(1) NaOH + HCl
(2) NaOH + CH_3COOH
(3) NH_4OH + HCl
(4) CH_3COOH + NH_4OH

69. A gas is allowed to expand in a well insulated container against a constant external pressure of 4.2 atm from initial volume of 1.25 L to the final volume of 6.75 L. The change in internal energy of the gas will be
(1) −4.21 kJ
(2) −2.5 kJ
(3) −8.1 kJ
(4) −1.82 kJ

70. In which of the following pairs of molecules/ions, the central atom is sp^3 hybridised?
(1) SO_3 and NO_3^-
(2) SO_4^{2-} and PO_4^{3-}
(3) SF_4 and CH_4
(4) PCl_3 and ClF_3

71. 2 mole of ideal gas at 27°C temperature is expanded reversibly from 2 L to 20 L. Entropy change in cal K⁻¹ (R = 2 cal/mol K) for the process is
(1) 92.1
(2) 0
(3) 4
(4) 9.2

@examix

72. The energy required to raise the temperature of 50 g of Cu from 35°C to 80°C
(Specific heat of copper is 0.39 $Jg^{-1}\,°C^{-1}$)
(1) 877.5 J
(2) 438.75 J
(3) 219.375 J
(4) 785.6 J

73. Which among the following has highest lattice enthalpy?
(1) NaI
(2) NaBr
(3) NaCl
(4) NaF

74. Number of bonding and antibonding electrons present in O_2^- respectively are
(1) 8 and 8
(2) 10 and 5
(3) 8 and 7
(4) 12 and 3

75. Given below are two statements: one is labelled as **Assertion (A)** and other is labelled as **Reason (R)**
Assertion (A): The resultant dipole moment of NH_3 is greater than that of NF_3
Reason (R): In case of NH_3, the orbital dipole due to lone pair is in the same direction as the resultant dipole moment of N-H bonds whereas orbital dipole is in opposite direction to that of N-F bonds in NF_3
In the light of above statements, choose the **correct** option.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) (A) is false but (R) is true

76. For which of the following reaction $\Delta H \neq \Delta U$?
(1) $H_2(g) + Cl_2(g) \to 2HCl(g)$
(2) $N_2(g) + O_2(g) \to 2NO(g)$
(3) $C(s) + O_2(g) \to CO_2(g)$
(4) $CH_4(g) + 2O_2(g) \to CO_2(g) + 2H_2O(l)$

77. If the enthalpy of hydrogenation of cyclohexene and benzene are −x kJ mol^{-1} and −y kJ mol^{-1} respectively then resonance energy of benzene will be
(1) 2x − y
(2) y − x
(3) y − 3x
(4) x − y

78. Formal charge on the central oxygen atom in O_3 molecule is
(1) Zero
(2) −1
(3) +1
(4) −2

9. Consider the following statements
Statement I: The energy of σ $2p_z$ molecular orbital is higher than that of π $2p_x$ molecular orbital for F_2
Statement II: The energy of σ* $2p_z$ molecular orbital is higher than that of π* $2p_x$ molecular orbital for N_2
In the light of above statements, choose the correct option.
(1) Both statement I and statement II are correct
(2) Both statement I and statement II are incorrect
(3) Statement I is incorrect while statement II is correct
(4) Statement I is correct while statement II is incorrect

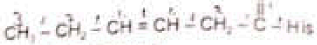
80. An example of odd electron molecule is
(1) F_2
(2) BCl_3
(3) NO_2
(4) O_3

81. Correct order of stability is
(1) $N_2^- > N_2^+ > N_2$
(2) $N_2 > N_2^+ > N_2^-$
(3) $N_2 > N_2^- > N_2^+$
(4) $N_2^+ > N_2 > N_2^-$

82. The species which does not exist is
(1) C_2
(2) Be_2
(3) N_2
(4) H_2^+

83. If internuclear axis is z, then zero overlap will take place between
(1) $2p_x$ and $2p_x$ orbitals
(2) 2s and $2p_z$ orbitals
(3) $2p_x$ and $2p_y$ orbitals
(4) $2p_y$ and $2p_y$ orbitals

84. The total number of σ and π bonds in the molecule are


$CH_3-CH_2-CH=CH-CH_2-\overset{\overset{O}{\parallel}}{C}-H$ is

(1) 14 σ bonds and 2 π bonds
(2) 14 σ bonds and 1 π bond
(3) 16 σ bonds and 3 π bonds
(4) 16 σ bonds and 2 π bonds

85. For which pair of the species, the standard molar enthalpies of formation at 298 K is zero?
(1) $Br_2(l)$ and $H_2(g)$
(2) $H_2(g)$ and $Cl_2(g)$
(3) C(diamond) and S(rhombic)
(4) S(g) and HCl(g)

86. Standard entropies of X_2, Y_2 and XY_3 are 60, 40 and 50 J K^{-1} mol^{-1} respectively. If the reaction $\frac{1}{2}X_2(g) + \frac{3}{2}Y_2(g) \to XY_3(g)$, $\Delta H = -30\,kJ$ to be at equilibrium the temperature should be
(1) 500 K
(2) 650 K
(3) 800 K
(4) 1000 K

87. The correct order of increasing bond order of C–O bond among CO, CO_2 and CO_3^{2-} is
(1) $CO < CO_2 < CO_3^{2-}$
(2) $CO_2 < CO < CO_3^{2-}$
(3) $CO_3^{2-} < CO < CO_2$
(4) $CO_3^{2-} < CO_2 < CO$

88. 10 moles of an ideal gas expands isothermally from 5 atm to 2 atm at 298 K. The change in internal energy for the process is
(1) Zero
(2) 22.8 KJ mol^{-1}
(3) 30.33 KJ mol^{-1}
(4) 101.5 KJ mol^{-1}

89. The correct order of bond order of O_2, O_2^+, O_2^-, O_2^{2-} is
(1) $O_2^{2-} > O_2^- > O_2 > O_2^+$
(2) $O_2^{2-} > O_2 > O_2^+ > O_2^-$
(3) $O_2^+ > O_2 > O_2^- > O_2^{2-}$
(4) $O_2 > O_2^+ > O_2^- > O_2^{2-}$

90. A reaction will be non-spontaneous at all temperatures if
(1) ΔH = +ve and ΔS = +ve
(2) ΔH = −ve and ΔS = −ve
(3) ΔH = −ve and ΔS = +ve
(4) ΔH = +ve and ΔS = −ve

BOTANY

y florets of sunflower have
(1) Hypogynous flower
(2) Inferior ovary
(3) Superior ovary
(4) Half inferior ovary

92. Given below are two statements:
Statement I: Both floral formula and floral diagram show placentation and union of carpels in gynoecium.
Statement II: Diadelphous stamens (9 + 1) are seen in pea.
In the light of the above statements, choose the **correct** answer from the options given below
(1) Statement I is incorrect but Statement II is correct
(2) Both Statement I and Statement II are correct
(3) Both Statement I and Statement II are incorrect
(4) Statement I is correct but Statement II is incorrect

93. The leaflets are attached at a common point, i.e., at the tip of the petiole in
(1) Neem
(2) Guava
(3) Calotropis
(4) Silk cotton

94. Match List I with List II

List I	List II
A. [illegible]	(I) Twisted aestivation
B. [illegible]	(II) Papilionaceous flower
C. [illegible]	(III) [illegible]
D. [illegible]	(IV) Marginal placentation

Choose the **correct** answer from the options given below
(1) [illegible]
(2) A-I, B-II, C-III, D-IV
(3) [illegible]
(4) A-II, B-III, C-IV, D-I

95. If a fruit is formed without fertilisation of the ovary then it is called
(1) Aggregate fruit
(2) False fruit
(3) Parthenocarpic fruit
(4) True fruit

96. In a bean flower, the largest petal which overlaps the two lateral petals is known as
(1) Keel
(2) Wings
(3) Standard
(4) Canna

97. Match the column-I with column-II and select the **correct** option

	Column-I		Column-II
a.	Staminode	(i)	Petals united
b.	Syncarpous	(ii)	Sterile stamen
c.	Gamopetalous	(iii)	Sepals free
d.	Polysepalous	(iv)	Fused carpels

(1) a(iv), b(ii), c(i), d(iii)
(2) a(ii), b(i), c(iv), d(iii)
(3) a(ii), b(iv), c(i), d(iii)
(4) a(i), b(ii), c(iii), d(iv)

98. Epipetalous stamens are present in
(1) Lily
(2) China rose
(3) Belladonna
(4) Pea

99. The fruit having fibrous mesocarp, which is used in making coir is
(1) Mango
(2) Apple
(3) Coconut
(4) Jackfruit

100. Select the **incorrect** match

(1)	Leaf base	-	It may bear two lateral small leaf like structures
(2)	Pulvinus	-	Swollen leaf base in some leguminous plants
(3)	Leaf blade	-	In monocots, it expands into a sheath covering the stem partially or wholly
(4)	Petiole	-	When it is long, thin and flexible, it allows leaf blades to flutter in wind

(1) (1)
(2) (2)
(3) (3)
(4) (4)

101. In a compound leaf
(1) Each leaflet has axillary bud
(2) Rachis is midrib
(3) There is parallel venation only
(4) Petiole is always absent

102. Select the **odd** one w.r.t. symmetry of flower.
(1) Gulmohur
(2) Cassia
(3) Chilli
(4) Bean

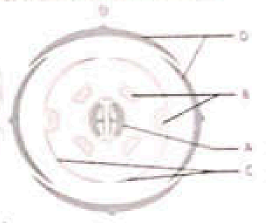
103. Choose the **incorrect** one for the cells present a few millimetres just above the root cap.
(1) Small in size
(2) Show rapid elongation
(3) Thin walled
(4) Presence of dense protoplasm

104. The arrangement of flowers on the floral axis of the plants known as
(1) Placentation
(2) Inflorescence
(3) Phyllotaxy
(4) Aestivation

105. Monoadelphous stamens are found in
(1) Citrus
(2) Mustard
(3) Pea
(4) China rose

106. Which among the following is absent in castor seed?
(1) Scutellum
(2) Endosperm
(3) Hilum
(4) Testa

107. Figure shows the floral diagram of the member of a family with labels A to D. Select the option which gives correct identification of label and its characteristic.



(1) B Stamens – Show variation in the length of filaments within a flower
(2) A Ovule – Axile placentation
(3) C Petals – Imbricate aestivation
(4) D Sepals – Valvate aestivation

108. A virus differs from a viroid as it
(1) Contains a nucleic acid
(2) Has proteinaceous capsid
(3) Is smaller than viroids
(4) Does not cause diseases in plants

109. In brinjal
(1) Gynoecium is situated in the centre and other floral whorls are present at the same level.
(2) Margins of thalamus grow and enclose ovary completely
(3) Gynoecium occupies the highest position while other floral whorls are situated below it
(4) Half inferior ovary is present

@examlix

110. Br symbol in floral formula is used to denote
(1) Fusion between floral parts
(2) Adhesion in floral parts
(3) Bracteate flower
(4) Ebracteate flower

111. Lichens are considered very good pollution indicators as they
(1) Grow well in polluted areas
(2) Grow only when the SO_2 level in atmosphere is high
(3) Become green when level of pollution increases
(4) Do not grow in polluted areas

112. Which of the following is usually **not** the function of root?
(1) Synthesis of plant growth regulators
(2) Absorption of water and minerals from soil
(3) Storage of reserve food materials
(4) Manufacturing of food and its transportation to aerial parts of plant

113. Read the following statements and state them as **true(T)** or **false(F)** and select the **correct** option.
a. Anthers are usually bilobed and each lobe contains two pollen sacs.
b. Corolla is the second whorl of the flower which is generally brightly coloured and have fragrance.
c. Sepals are generally green leaf-like structure that protect the flower in bud stage.
(1) a-(T), b-(F), c-(F)
(2) a-(F), b-(T), c-(T)
(3) a-(T), b-(T), c-(T)
(4) a-(F), b-(T), c-(F)

114. In wheat plant
(1) Fibrous roots directly elongate from radicle
(2) Primary root is short lived
(3) Tap root system is present
(4) Roots arise from the upper nodes of the stem

115. Papilionaceous corolla is seen in
(1) Gram
(2) Tomato
(3) Chilli
(4) Lily

116. Select the **correctly** matched pair
(1) Lotus – Has syncarpous gynoecium
(2) *Alstonia* – The leaflets are attached at tip of the petiole
(3) *Monstera* – Roots arise from parts of plant other than radicle
(4) Canna – Actinomorphic symmetry is observed in its flowers

17. Dinoflagellates are mostly
(1) Holozoic and motile
(2) Multinucleated and lack cell wall
(3) Marine and photosynthetic
(4) Colourless and lack motility

8. Perianth is a term for
(1) No distinction in calyx and Corolla
(2) No distinction between androecium and Gynoecium
(3) Coloured sepals
(4) Colourless petals

119. Match the following columns and select the **correct** option.

	Column I		**Column II**
(a)	T.O. Dener	(i)	Discovered infectious agent that was smaller than viruses
(b)	Dmitri Ivanovsky	(ii)	Showed that viruses could be crystallised and crystals consist largely of proteins
(c)	M.W. Beijerinck	(iii)	Recognised certain microbes as causal organism of the mosaic disease of tobacco
(d)	W.M. Stanley	(iv)	Called the infectious fluid as Contagium vivum fluidum

(1) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)
(2) (a)-(i), (b)-(iii), (c)-(iv), (d)-(ii)
(3) (a)-(iv), (b)-(ii), (c)-(iii), (d)-(i)
(4) (a)-(ii), (b)-(iii), (c)-(i), (d)-(iv)

120. Bacterial viruses usually contain
(1) Double stranded RNA
(2) Single stranded RNA
(3) Double stranded DNA
(4) Single stranded DNA

121. Creutzfeldt Jacob disease in humans is caused by
(1) A virus that also causes the mad cow disease in cattle
(2) An infectious agent that is similar in size to the causative agent of citrus canker
(3) A pathogen that was found to be a free RNA
(4) An analogous variant of the infectious agent that causes bovine spongiform encephalopathy in cattle

122. Identify the following statements as True **(T)** or False **(F)** and select the **correct** option.
A. In lichens, the fungal component is known as mycobiont.
B. Bladderwort, Venus fly trap and *Cuscuta* are insectivorous plants.
C. In imperfect fungi, reproduction occurs only by sexual spores known as oospores.
D. Plant affected by viruses can have symptoms like leaf rolling and curling.
(1) A-(T), B-(F), C-(T), D-(T)
(2) A-(F), B-(T), C-(F), D-(F)
(3) A-(T), B-(T), C-(F), D-(T)
(4) A-(T), B-(F), C-(F), D-(T)

123. Read the following assertion(A) and reason (R) and select the **correct** option.
Assertion (A): In slime moulds, during unfavourable condition, plasmodium differentiates and form fruiting bodies bearing spores.
Reason (R): The spores possess true walls and are extremely resistant which help slime mould to survive for many years.
(1) Only (A) is true
(2) Both (A) and (R) are false
(3) Both (A) and (R) are true and (R) is the correct explanation of (A)
(4) Both (A) and (R) are true but (R) is not the correct explanation of (A)

124. Which of the following is **not** edible fungi?
(1) Yeast
(2) *Puccinia*
(3) Morels
(4) Truffels

125. Mycelium is aseptate and coenocytic in
(1) *Penicillium*
(2) *Albugo*
(3) *Agaricus*
(4) *Aspergillus*

126. Read the following statements and select the **correct** option.
Statement A: All protozoans are heterotrophs.
Statement B: Protozoans are believed to be primitive relatives of animals.
(1) Only statement A is correct
(2) Only statement B is correct
(3) Both statements are incorrect
(4) Both statements are correct

127. Red tide is caused by
(1) Desmids
(2) Diatoms
(3) *Gonyaulax*
(4) A cyanobacterium

128. Which among the following characteristics is **not** true for the organism which are considered as the chief producers in the ocean?
(1) They float passively in water currents
(2) Their cell wall is embedded with silica which makes it indestructible
(3) They are not found in the fresh water
(4) They have the photosynthetic pigments such as chlorophyll

129. Euglena has
(1) Pellicle
(2) Cell wall
(3) Cilia
(4) Only a single flagellum

130. Mark the **correct** match
(1) Cyanobacteria – Absence of flagella throughout the life cycle
(2) Chrysophytes – Some marine forms possess silica shells
(3) Sporozoans – Have a definite region of ingestion (gullet) and egestion
(4) Mycoplasma – Infects only animals

131. Match column I with column II and select the correct option.

	Column I		Column II
a.	Albugo	(i)	Fungi extensively used in biochemical and genetic work
b.	Neurospora	(ii)	Fungi causing smut
c.	Ustilago	(iii)	The parasitic fungi on mustard
d.	Alternaria	(iv)	Fungi which show only vegetative phase

(1) [illegible]
(2) a(iii), b(i), c(ii), d(iv)
(3) [illegible]
(4) [illegible]

132. The causative agent of the potato spindle tuber disease is
(1) A nucleoprotein
(2) Infectious RNA particle with low molecular weight
(3) A symbiotic association between algae and fungi
(4) An abnormally folded protein

133. Select the pair of fungi which belongs to class Deuteromycetes
(1) Alternaria and Morels
(2) Aspergillus and Bracket fungi
(3) Colletotrichum and Alternaria
(4) Trichoderma and Aspergillus

134. Vegetative reproduction in fungi occurs by all **except**
(1) Fission
(2) Budding
(3) Fragmentation
(4) Spore formation

135. In sac fungi, the site of karyogamy and meiosis is
(1) Oogonium
(2) Ascus
(3) Zygote
(4) Basidium

ZOOLOGY

136. The most important primary factor in binding of haemoglobin with oxygen is
(1) Partial pressure of oxygen
(2) Acidity
(3) Partial pressure of carbon dioxide
(4) Temperature

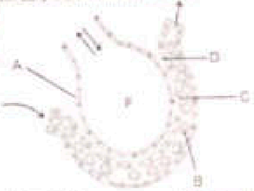
137. Oxygen dissociation curve is plotted with __A__ on Y-axis and __B__ on X-axis to obtain a sigmoid curve. A and B respectively are
(1) % saturation of Hb with O_2 and pO_2
(2) % saturation of Hb with CO_2 and pO_2
(3) pO_2 and % saturation of Hb with O_2
(4) pCO_2 and % saturation of Hb with O_2

138. The partial pressure of O_2 in systemic veins carrying deoxygenated blood is equal to
(1) pCO_2 in alveolus
(2) pCO_2 in tissues
(3) pO_2 in alveolus
(4) pCO_2 in systemic veins

139. Which of the following conditions are found in alveoli of lungs?
(1) High pO_2, low pCO_2 and low H^+
(2) High pH, high pCO_2 and high pO_2
(3) High H^+, low pO_2 and high pCO_2
(4) Low pH, high pCO_2 and low pO_2

140. In humans, 70% of CO_2 is transported
(1) As oxyhaemoglobin
(2) In bicarbonate form
(3) In dissolved form in plasma
(4) In association with haemoglobin

141. Observe the figure given below and select the **correct** option that comprises the diffusion membrane



(1) A, B and C
(2) B, C and D
(3) A, C and D
(4) C, D and E

142. Given below are the features of a respiratory regulatory centre called 'X'
(a) Neural signals from this centre can reduce the duration of inspiration
(b) It can moderate the functions of respiratory rhythm centre
(c) It is present in pons
Select the option that correctly identifies 'X'
(1) Pneumotaxic centre
(2) Hunger centre
(3) Chemosensitive area
(4) Association area

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143. Read the statements A and B and choose the **correct** option.
Statement A : Solubility of the gases as well as the thickness of the diffusion membrane involved affect the rate of diffusion.
Statement B : Pressure contributed by an individual gas in a mixture of gases is called partial pressure.
(1) Both statements A and B are incorrect
(2) Statement A is correct but statement B is incorrect
(3) Both statements A and statement B are correct
(4) Statement B is correct but statement A is incorrect

144. Which among the following is **not** true for the equation given below?

$$CO_2 + H_2O \overset{(1)}{\rightleftharpoons} H_2CO_3 \overset{(2)}{\rightleftharpoons} HCO_3^- + H^+$$

(1) Reaction (1) is catalyzed by carbonic anhydrase
(2) Reaction (1) primarily occurs in plasma
(3) Reaction (2) is catalyzed by a class IV enzyme
(4) Reaction (2) results in lowering of blood pH upon holding breath

145. Which of the following is the most common feature of occupational respiratory disorders?
(1) Collapsing of vascularised alveoli
(2) Fibrosis of lungs
(3) Difficulty in breathing causing wheezing due to bronchodilation
(4) Fibrosis of oesophagus

146. Under normal physiological conditions, how much carbon dioxide is delivered by 5 L of deoxygenated blood to the alveoli?
(1) 200 mL (2) 400 mL
(3) 100 mL (4) 300 mL

147. Peripheral chemoreceptors that help in regulation of breathing are present in
(a) Aortic arch
(b) Carotid artery
(c) Medulla oblongata
Choose the **correct** option
(1) (a) only (2) (b) only
(3) (a) and (b) only (4) (a) and (c) only

148. Which of the following factors favour decrease in breathing rate back to normal levels?
(1) Increased blood oxygen levels
(2) Increase blood CO_2 levels
(3) Increased metabolism
(4) Increase in acidity of body fluid

149. Exchange of gases between lungs and pulmonary capillaries takes place by/with
(1) Simple diffusion (2) Facilitated diffusion
(3) Active process (4) The help of proteins

150. Which of the following factors do **not** favour the formation of oxyhaemoglobin?
(1) High pCO_2 and low pH
(2) High pO_2 and high pH
(3) Lesser H^+ concentration and lower temperature
(4) Low pCO_2 and low H^+ concentration

151. Read the following statements and choose the correct option.
Statement A : In emphysema, the surface area of gaseous exchange is reduced.
Statement B : Due to excessive smoking, the walls of alveoli are damaged.
(1) Both statements are incorrect
(2) Both statements are correct
(3) Only statement A is incorrect
(4) Only statement B is incorrect

152. Structure in lungs which is directly involved in exchange of O_2 and CO_2 between air and blood in humans is
(1) Bronchi (2) Trachea
(3) Alveoli (4) Nasal chambers

153. Chemosensitive area is highly sensitive to
(1) CO_2 and H^+ (2) CO_2 and O_2 only
(3) O_2 and H^+ only (4) O_2 and temperature only

154. Complete the analogy by selecting the **correct** option.
Cardiac arrest : When heart stops beating :: Congestive heart failure : ______
(1) State of heart when it is not pumping blood effectively enough to meet the needs of the body
(2) Not enough oxygen reaching the heart muscles
(3) Heart muscle suddenly damaged by an inadequate blood supply
(4) Prolonged high blood pressure of 120/80 mmHg

155. Choose the **incorrect** statement w.r.t blood
(1) Blood is a fluid connective tissue
(2) It consists of formed elements and plasma
(3) Blood cells and plasma both are responsible for transportation of O_2 and CO_2
(4) Cells of blood form matrix and structural proteins like other connective tissues

156. What is the graveyard of RBCs in humans?
(1) Stomach (2) Spleen
(3) Liver (4) Lymph nodes

157. Match the column I with column II and choose the **correct** option w.r.t an adult healthy human.

	Column I		Column II
a.	Neutrophils	(i)	2-3% of total WBCs
b.	Basophils	(ii)	6-8% of total WBCs
c.	Monocytes	(iii)	20-25% of total WBCs
d.	Eosinophils	(iv)	60-65% of total WBCs
		(v)	0.5-1% of total WBCs

(1) a(ii), b(v), c(iii), d(i) (3) a(iii), b(ii), c(iv), d(v)
(2) a(iv), b(v), c(ii), d(i) (4) a(iv), b(v), c(i), d(ii)

158. Individuals called 'universal donors' belong to the blood group
(1) A (2) AB
(3) O (4) B

159. Choose the **incorrect** statement.
(1) Lymph lacks haemoglobin but contains specialised lymphocytes
(2) Lymph has same mineral distribution as that of plasma
(3) The lymphatic system collects the fluid and drains it back directly into the major arteries
(4) Lymph acts as a transport fluid for nutrients, hormones etc.

160. During coagulation of blood the conversion of inactive fibrinogen into fibrin is catalysed by
(1) Thromboplastin (2) Prothrombin
(3) Thrombokinase (4) Thrombin

161. Choose the **incorrect** statement.
(1) RBCs have an average life span of 120 days
(2) A healthy man has 12-16 gms of haemoglobin in every 100 ml of blood
(3) Leucocytes are colourless due to lack of haemoglobin
(4) Plasma with clotting factor is called serum

162. The condition called erythroblastosis foetalis can be avoided by
(1) Injecting Rh antigen into mother's blood immediately after delivery of the first child
(2) Administering anti-Rh antibodies to the newborn
(3) Administering anti-Rh antibodies to mother immediately after delivery of first Rh positive child
(4) Injecting Rh antigen into mother's blood just after conception

163. The first heart sound occurs due to
(a) Closure of tricuspid valve
(b) Opening of semilunar valve
(c) Closure of bicuspid valve
(d) Closure of semilunar valves
Choose the **correct** option
(1) (a) and (b) (2) (b) and (c)
(3) (b) only (4) (a) and (c)

164. The formed elements in blood produced from megakaryocytes are
(1) Erythrocytes (2) Thrombocytes
(3) Leucocytes (4) Neutrophils

165. Select the **incorrect** match.
(1) Mitral valve – Between left atrium and left ventricle
(2) Bicuspid valve – Between left ventricle and aorta
(3) Tricuspid valve – Between right atrium and right ventricle
(4) Semilunar valve – Between right ventricle and pulmonary artery

166. Which of the following is mainly involved in maintaining the osmotic balance of body fluids?
(1) Fibrinogen (2) Globulin
(3) Thrombokinase (4) Albumin

167. **Assertion (A)** : On activation of parasympathetic nervous system, cardiac output decreases.
Reason (R) : Parasympathetic nerve endings release adrenaline which decreases rate of heart beat and speed of conduction of action potential.
In the light of above statements, select the **correct** option.
(1) Both Assertion and Reason are false statements
(2) Assertion is true statement but Reason is false
(3) Both Assertion & Reason are true but reason is not the correct explanation of assertion
(4) Both Assertion & Reason are true and the reason is the correct explanation of assertion

168. Closed circulatory system with 4 chambered heart is found in all of the following, **except**
(1) Birds (2) Mammals
(3) Amphibians (4) Crocodiles

169. All of the following structures are involved in systemic circulation, **except**
(1) Dorsal aorta, capillary (2) Capillary, vena cava
(3) Right atrium, left ventricle (4) Left atrium, right ventricle

170. T wave in electrocardiogram represents
(1) The return of ventricles from excited to normal state
(2) Depolarisation of ventricles from normal to excited state
(3) Onset of atrial systole
(4) Onset of atrial diastole

171. Select the **incorrect** statement among the following
(1) During ventricular systole, tricuspid and bicuspid valves get closed
(2) During joint diastole, tricuspid and bicuspid valves open
(3) During ventricular diastole, semilunar valves open
(4) The SAN generates an action potential which stimulates both atria causing atrial systole

172. Maximum ventricular filling occurs when
(1) Ventricles relax and atria contract
(2) Ventricles contract and atria relax
(3) Both ventricles and atria contract
(4) Both ventricles and atria relax

173. Pulmonary veins from lungs are known for carrying __A__ blood to the __B__ of heart. Choose the option that fills A and B correctly.

	A	B
(1)	Oxygenated	Right atrium
(2)	Deoxygenated	Right ventricle
(3)	Oxygenated	Left atrium
(4)	Deoxygenated	Left ventricle

(1) (1) (2) (2)
(3) (3) (4) (4)

174. In a mm^3 of blood, the maximum number and minimum number of cells would be respectively
(1) WBCs, RBCs (2) RBCs, WBCs
(3) RBCs, Platelets (4) Platelets, WBCs

175. Identify and select the option representing the conducting part of cardiac musculature that is connected with AV node and passes through the interventricular septum
(1) Bundle of His (2) Purkinje fibre
(3) Chordae tendineae (4) Sino-auricular node

176. If the cardiac output of a person is 4900 mL and blood volume in the left ventricle at the end of diastole is 140 mL while at the end of ventricular systole is 70 mL, then what is the heart rate of the person?
(1) 75 beats per minute (2) 70 beats per minute
(3) 35 beats per minute (4) 140 beats per minute

177. The blood flows freely into the ventricles during all the following events **except**
(1) Atrial systole (2) Ventricular systole
(3) Ventricular diastole (4) Joint diastole

178. The middle layer of coronary artery is composed of
(1) Uninucleated, fusiform and voluntary muscle fibres alongwith collagen fibres
(2) Multinucleated, cylindrical and voluntary muscle fibres alongwith elastic fibres
(3) Uninucleated, fusiform and involuntary muscle fibres alongwith elastic fibres
(4) Multinucleated, cylindrical and involuntary muscle fibres alongwith collagen fibres

179. The atrium and the ventricle of the same side are separated by atrio-ventricular septum made up of a
(1) Thin muscular wall (3) Thick fibrous tissue
(2) Thin calcified rigid wall (4) Thick elastic cartilage

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180. Which blood cells are responsible for secreting heparin, histamine and serotonin?

(1) Eosinophils [✓ (2) Basophils]

(3) Neutrophils (4) Monocytes

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